CH 3 - Preference

Luke Lu • 2026-02-25

Consumer Preference Model

🐳 **Info — Consumer Preference Model**

- Model of what and how mohch consumers purchase
- Model of a choice between alternative

Effect on the model

- Price
- Preference
  - Weak axiom of reveald preference: if choosing between A, B, if one of the two becomes unavailable. The remaining choice must contain A to remain a rational choice.
- Income/Budget (next page)

🐳 **Info — Budget**

1. Equation
   - $M =$ income
   - $p_i =$ price for product i
   - $q_i =$ quantity for product i

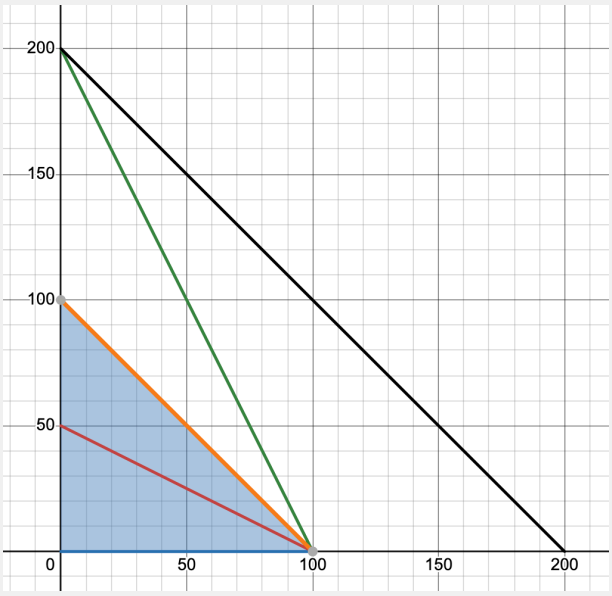
   $$M = p_1 q_1 + p_2 q_2$$

   $\longrightarrow M$ is the amount of money allocated for this specific decision

   Note that the goal is to gain the most amount of happiness from the given budget.

   Example allocation:
   - Salary
     - Spending
       - Shopping
     - Saving

2. Graph



   Area below the curve is the Inefficient distribution of budget.

   On the line suggests the optimial distribution of budget.

   The slope $\frac{p_2}{p_1}$: good 2 costs $\frac{p_2}{p_1}$ good 1 $\Longrightarrow$ opportunity costs

   Orange: original line
   Green: p_2 decreases
   Red: p_2 increases
   Black: M increases

🐳 **Info — Preferences**

Preference often/always intangible and non-numeric.

Force intangible to have inherent structure like numbers

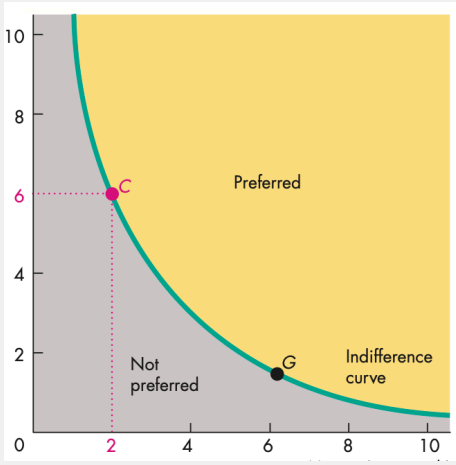
"Happiness" is equivalent "Utility"

An assumption is that of deminishing marginal utility, but still positive.

Marginal rate of Substitution(MRS) : The rate of which the consumer would trade between q_1, q_2

Due to the continuously deminishing MRS (continuously deminishing rate of trade), the curve will not be linear like the budget(constant rate of trade).

Graph

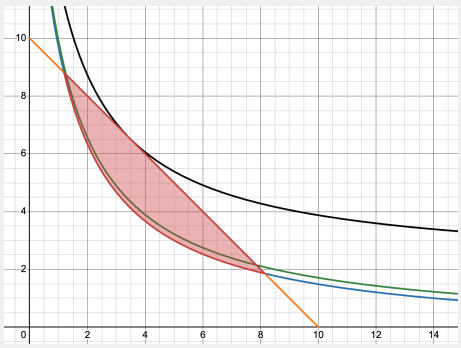


- Every point above the curve is better, more utility: (i.e. yellow area)
- Every point below the curve is worse, less utility: (i.e. grey area)
- Every point on the curve is equivalent to marginal utility
- The green curve is also called **Indifference Curve**

🐳 **Info — Utility length**

Length of opportunity: the area enclosed by the utility curve and the budget curve. (i.e. red area)

There always exists another parallel utility curve that is better than the previous one.



🐳 **Info — Relationships between the curves**

The budget line becomes a tangent line of the utility curve at exactly the optimal point.

MRS = slope of the budget curve = Ratio of prices = $\frac{M_{u_1}}{M_{u_2}} = \frac{p_1}{p_2} \implies \frac{M_{u_1}}{p_1} = \frac{M_{u_2}}{p_2}$

The marginial utility per dollar is equivalent at the optimal point

If $\frac{M_{u_1}}{p_1} > \frac{M_{u_2}}{p_2}$, then item 1 brings more utility.

- Buy item 1, thus M_{u_1} decreases
- To afford item 1, sell item 2, and M_{u_2} increases

Since MRS is deminishing, thus explaining the price of luxury goods are hight as essentials have lower MRS as they are deminishing since we have obtained enough of it.

Example:

Given

$P_1 = 10, P_2 = 5, M = 40$

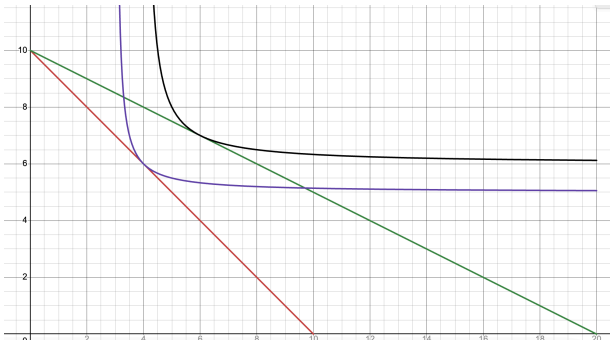
Item 1	M_u	$\frac{M_u}{P}$
1	40	4
2	20	2
3	10	1
4	5	0.5

Item 2	M_u	$\frac{M_u}{P}$
1	20	4
2	15	3
3	10	2
4	5	1

Table	Item	Cost	Gain
1	1	10	40
2	2	5	20
3	2	5	15
4	1	10	20
5	2	5	10
6	2	5	5
Total		$40	110 Utility

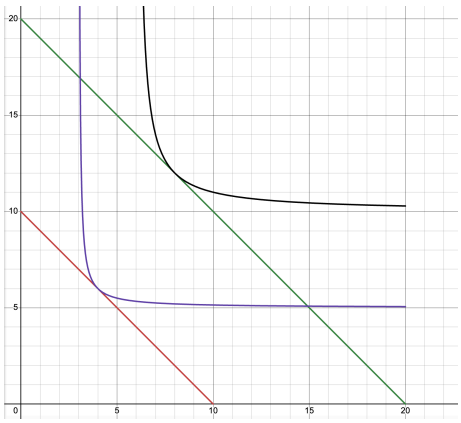
What happens if the price of Item 1 decreases by half: $P_1 = 5$

Red: initial curve
Green: new curve
Purple: initial utility curve
Black: new utility curve



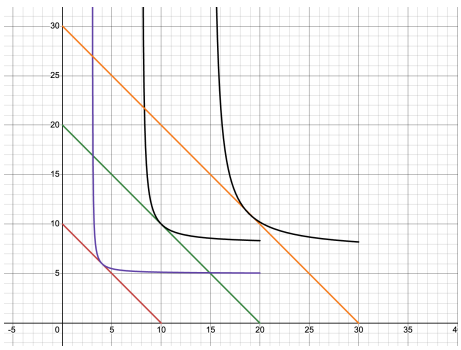
What if the income rises

Red: initial curve
Green: new curve
Purple: initial utility curve
Black: new utility curve

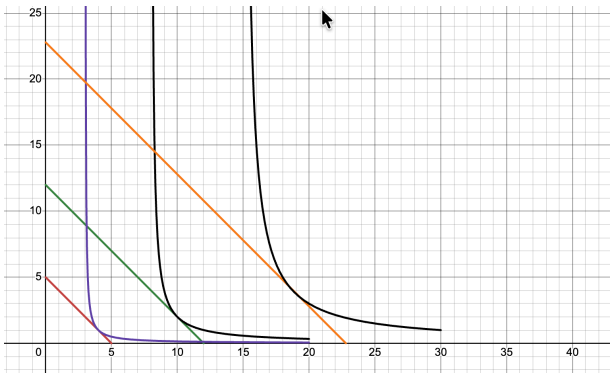


How could depict inferior goods (i.e. instant ramen)

Red: initial curve
Green: new curve
Purple: initial utility curve
Black: new utility curve Orange: prediction on possible new curve



What if curves lean but at constant rate?

For one extreme, the other extreme is the higher, that is the line passes through the intersection of the utility curve and budget line have slope > 1

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